The Kerekes Handshake

Evidence Anchoring Framework
Whitepaper & Full Specification — v1.6.1

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The honest position, stated first: This protocol does not prevent fraud. Self-certification is self-lying. A motivated actor can fabricate evidence and sign it with PGP. What this protocol does is raise the cost and detectability of lying by anchoring claims to inspectable artifacts and, where possible, linking them to independent public records. The goal is a better signal-to-noise ratio than the current system — which, for self-asserted professional claims, has essentially none. That is a worthwhile improvement. It is not a solved problem.

1. Origin

This protocol began as a personal frustration: after fifteen years of self-employed work across civic, clinical, and physical systems, writing a resume that fit the standard format meant compressing real evidence into two pages of narrative. The evidence existed — FOI rulings, inspection reports, audit spreadsheets, municipal records — but the format had no place for it.

The question was practical: is there a simple, open way to link a resume bullet to its primary source artifacts so that anyone — human or AI — can follow the chain? The answer turned out to be yes, and the approach turned out to generalize beyond resumes.

This document describes what was built, how it works, where it is genuinely useful, and where its limits are.

> *The reference implementation uses a real career as its dataset. When you fork this for your own use, substitute your own evidence. Real artifacts are always preferable to synthetic ones.*

2. The Problem

The Keyword Genius Paradox

AI tools can generate a polished, keyword-optimized resume for anyone in seconds. When every candidate is a keyword genius, the signal-to-noise ratio of resume screening drops toward zero. The same dynamic applies to product marketing copy, political bios, and nonprofit impact statements. Narrative optimization has become nearly free.

The consequence is not that everyone lies — most people don't. The consequence is that the format can no longer distinguish those who did the work from those who described it well. Both look identical to a parser.

The Self-Certification Problem

The traditional response to false claims is third-party verification: a credential issuer, a background check service, a reference call. These work reasonably well for standardized credentials — degrees, licenses, certifications. They fail completely for the class of accomplishments that no institution will ever certify: a municipal budget reconstruction, a volunteer fire department compliance sprint, a solo residential build. These things happened. The evidence exists. But there is no issuer.

Self-certification is the only option — which means, structurally, self-lying is equally easy. Apostille systems, notarization chains, and institutional verification frameworks took centuries to develop precisely because humans always lied and the cost of that lying was high enough to justify the infrastructure. That infrastructure works by adding independent vouchers at each step. The Kerekes Handshake does not replicate that chain. It does something more modest: it makes the evidence inspectable, and links it to independent public records where they exist.

3. The Approach

The protocol has one core idea: every claim should be accompanied by the primary artifacts that support it, structured so that any AI agent or human auditor can retrieve and inspect them without special tools or platform access.

This is implemented through three layers:

```
Narrative Layer → Registry Layer → Integrity Layer
```

Each layer serves a different audience: Narrative for humans, Registry for machines, Integrity for auditors verifying the chain has not been tampered with.

- **Layer 1 — Narrative (HTML/PDF):** The human-readable summary. Claims are tagged with `data-kcm` attributes that map each claim to a unique ID in the registry.
- **Layer 2 — Registry (JSON):** The `claims.json` ledger. Maps each claim ID to one or more absolute evidence URIs, an outcome description, and an optional `external_verification` link to a public record.
- **Layer 3 — Integrity (PGP):** The `site_manifest.json.asc` — a PGP-signed manifest containing SHA-256 hashes of all artifacts. Proves who vouched for the vault and when. Does not prove the underlying documents are authentic.

4. File Structure

```
/resume.html (or index.html)  <- Narrative Layer (data-kcm tags)
/claims.json                  <- Registry Layer (claim → evidence map)
```

```
/site_manifest.json.asc       <- Integrity Layer (PGP-signed hashes)
/evidence/                    <- Artifact vault (PDFs + .txt sidecars)
/query/                       <- AI audit portal
/openapi.yaml                 <- Machine-readable API schema
/llms-full.txt                <- LLM pre-flight context
/.well-known/claims.json      <- Universal discovery endpoint
```

All files are static. No database, no server-side code, no platform dependency. Any static host works: GitHub Pages, Netlify, Cloudflare Pages.

5. The Text Bridge

Born from real-world debugging, March 2026.

During testing, most AI crawlers failed to reliably parse binary PDF streams in real-time — stalling, skipping content, or hallucinating figures. The solution is simple: for every PDF artifact, maintain a matching plain-text sidecar with the same filename.

```
2011-main-spreadsheet-all.pdf
2011-main-spreadsheet-all.txt   <- same name, .txt extension
```

AI agents fall back to the `.txt` sidecar if the PDF is unreadable. This ensures accurate extraction of dates, numbers, and names. Generate sidecars automatically:

```
for f in *.pdf; do pdftotext "$f" "${f%.pdf}.txt"; done
```

6. KCM — Kerekes Claim Markup

KCM is a lightweight semantic HTML attribute layer. It requires no JavaScript, no external library, and no build step.

```
<article data-kcm="budget_audit_2011"
         aria-label="KCM Claim: budget_audit_2011"
         itemscope itemtype="https://schema.org/CreativeWork">
  <p itemprop="description">Municipal Budget Reconstruction</p>
  <footer class="evidence-grid">
    <a href="/evidence/audit.pdf" itemprop="citation">Audit [.pdf]</a>
    <a href="/evidence/audit.txt" itemprop="citation">Text Bridge [.txt]</a>
  </footer>
</article>
```

The `data-kcm` value must match a `claim_id` in `claims.json`. That is the only hard requirement. Everything else is optional enhancement.

7. claims.json — Field Reference

Field	Required	Description

claim_id	✓	Unique slug matching the data-kcm value
actor	✓	person | company | product | property | institution
claim	✓	Plain-language statement being asserted
evidence	✓	Array of URIs — include both .pdf and .txt versions
outcome	✓	Measurable or verifiable result
domain		System category (Civic, Human, Physical, Ecological)
external_verification		Links to .gov or third-party public records
verification_strength	✓ v1.6	Self-declared + AI-assessed score (see Section 10)
integrity		PGP fingerprint, signed_by, manifest URL

8. AI Integration

An AI agent with web access can traverse the full audit chain autonomously. The reference implementation has been tested against Grok (X.ai), which returned verified status on all six claims in the initial deployment and all nine claims in the current implementation.

- **Step 1:** Read `llms-full.txt` — pre-flight biographical context and audit instructions.
- **Step 2:** Fetch the resume HTML — read raw source to detect `data-kcm` attributes and the `KCM-CLAIMS` HTML comment.
- **Step 3:** Fetch `claims.json` — map claim IDs to evidence URIs and verification metadata.
- **Step 4:** Retrieve artifacts — use `.txt` sidecar if PDF is unreadable.
- **Step 5:** Verify SHA-256 hashes against `site_manifest.json.asc`.
- **Step 6:** Check `external_verification` links against public records where present.
- **Step 7:** Assess `verification_strength` independently on a 1–10 scale. Report: VERIFIED / CONFLICT / UNSTABLE per claim, with strength score and rationale.

9. The Vault Resume — Presentation Standard (v1.6)

The Vault Resume is an evidence-first presentation format for the Narrative Layer. It addresses a specific tension: recruiters scan resumes in roughly six seconds, while AI agents can audit evidence vaults in sixty. Traditional resume prose tries to serve both audiences and serves neither particularly well.

The approach is to separate them explicitly:

```
HUMAN LAYER (half-page):   Dense stubs -> six-second scan
                                          |
                                          v
AI LAYER (infinite vault):  Evidence retrieval -> audit -> verdict
```

Stub Format

Each claim is one dense line. No prose. No adjectives. The evidence qualifies the claim — the stub names it.

```
[Domain]: [intervention], [metric], [date] — REF: [claim_id]
```

Example from the reference implementation:

```
Civic: Municipal budget audit, $38k/household debt exposed,
       FIC 2009-014 secured, 2007-2011 — REF: budget_audit_2007_2011
```

What Belongs Where

On the page (human layer)	In the vault (AI layer)
Claim stub — one line	Full evidence PDFs + .txt sidecars
Key metric or outcome	Press coverage, archive records
Date range	Inspection reports, rulings, filings
REF: claim_id	Certifications, government documents
Audit link at top	PGP-signed manifest

10. Verification Strength — The Honest Layer

Self-certification is self-lying.

This protocol does not prevent fabricated evidence. A motivated actor can produce a forged permit scan and serve it with a valid PGP signature. The signature proves who vouched for the file and when — not that the underlying document is authentic.

Humans have been lying in professional contexts for as long as professional contexts have existed. The apostille chain — notary, Secretary of State, certified translator, consulate — took centuries to develop because people always lied and always will. That chain works by adding independent vouchers at each step. The Kerekes Handshake does not replicate it. What it does is raise the cost and detectability of lying. That is a meaningful improvement over the current baseline. It is not a guarantee.

The Verification Strength Spectrum

Not all evidence is equal. The `verification_strength` field in `claims.json` makes this explicit. Evidence quality ranges from self-authored documents (trivially fabricated) to live government verification endpoints (near-impossible to fake without state-level collusion).

Score	Evidence Type	Example	Forgery Difficulty
1–2	Self-authored, unlinked	"I saved $3.4M" — internal Word doc	Trivial
3–4	Third-party authored, no live endpoint	Press article (PDF only)	Moderate
5–6	Third-party + external_verification link	Press article + linked .gov URL	Hard
7–8	Government record with public lookup	FOI ruling, building permit	Very hard
9	Live government endpoint	License board lookup, congressional record	Near-impossible
10	Live QR-verified vital record	Italy-style embedded issuer verification	Impossible without state collusion

Italy and the Ceiling

Italy's vital records system embeds a QR code in the document itself that links directly to the issuing authority's verification endpoint. Scanning the QR confirms the document's authenticity in real time without contacting a third party. This represents the current practical ceiling of document verification for physical records. The `external_verification` field with `endpoint_type: live_lookup` is the Handshake's approximation of this for digital artifacts.

Self-Declared and AI-Assessed Scores

Every claim in v1.6 declares its verification strength through two mechanisms. The actor self-declares a score and provides a rationale. The auditing AI independently assesses the same scale after reviewing the evidence. Both scores are stored in `claims.json`:

```
"verification_strength": {
  "self_declared": 7,
  "rationale": "Permit linked to county building dept lookup.",
  "ai_assessed": null,
  "ai_rationale": null
}
```

A gap between `self_declared` and `ai_assessed` is itself a signal. An actor who claims 9 on a self-authored document and an AI that returns 2 tells the auditor something important — without requiring a binary pass/fail judgment.

What the Protocol Guarantees and Does Not

The protocol DOES	The protocol DOES NOT
Prove who vouched for the evidence	Prove the evidence is authentic
Make fabrication more expensive	Prevent fabrication
Surface conflicts with public records	Guarantee all public records are checked
Raise the cost of lying	Eliminate lying
Improve signal over noise	Eliminate noise

11. Universal Claims Standard (v1.6)

> *"Everyone is racing to put property deeds on the blockchain — but who is putting the new roof receipt in claims.json?"*

The protocol is not specific to resumes. The same `claims.json` structure, Text Bridge, and PGP manifest apply to any domain where a claim needs evidence. The universal circuit is identical:

```
Actor  ->  Claim  ->  Artifact Vault  ->  Verification
```

Use Cases and Honest Strength Assessments

Actor	Claim	Vault	Typical Strength
Job candidate	"Saved the city $3.4M"	FOI rulings, budget spreadsheets, press	7–9
Joe the Plumber	"Licensed for gas lines"	Permits, insurance certs, inspection photos	5–9
LG Dishwasher	"Cleans 30% better"	Lab data, Energy Star records, service manual	6–8
Property	"New roof (2022)"	Paid invoice, permit, inspection sign-off	7–9
Nonprofit	"90% to the field"	PGP-signed audits, IRS 990 public record	7–9
Politician	"Voted for the environment"	30-year roll-call + congress.gov	9
Designer	"Expert non-destructive editing"	.psd source file with full layer stack	3–5

Honest Limits by Domain

- **Professional credentials:** FOI rulings and government inspection records reach strength 7–9 when linked to live public endpoints. Self-authored documents are strength 1–2 regardless of how they are presented. The gap between a self-authored PDF and a verifiable government record is significant and should be declared honestly.
- **Trades:** Reaches strength 8–9 when state licensing databases are linked. Without a live endpoint, a license scan is strength 4–5 — better than nothing, but not independently verifiable.
- **Real estate:** Building permits are public record in most jurisdictions. A permit linked to a county database lookup is strength 7–8. A paid invoice with no external corroboration is strength 3–4. Disclose this honestly.
- **Civic/political:** Congressional and state legislative records are complete public data at strength 9. The protocol surfaces the record; it does not interpret it. The same voting history can support different conclusions depending on context. The Handshake provides the evidence, not the analysis.
- **Creative portfolios:** No government equivalent exists. Source files (PSD, AEP, WAV) are strength 3–5 — forensically richer than final outputs, but not independently verifiable against an external

record. Declare this honestly.

The Deterministic Collision

When `external_verification` links to a live government endpoint, fabrication becomes self-defeating. A plumber who provides a forged license scan linked to a state licensing database lookup will produce a collision the moment an AI agent checks the live record. No editorial judgment required — the records either match or they do not.

This is the protocol's strongest mechanism. It applies wherever a public record exists and is linkable. Where no public record exists, the protocol provides weaker but still useful evidence-anchored claims with honest strength declarations.

The Universal Evidence Endpoint

Any actor can host a `/.well-known/claims.json` endpoint as a standardized discovery point. A plumber hosts one on their business domain. A property developer hosts one per listing. A nonprofit hosts one alongside their annual report. This is how open standards spread: not through mandates, but through adoption.

12. Implementation Checklist

v1.5 Compliant

- **KCM Markup:** Resume HTML uses `data-kcm` on all claim containers.
- **Text Bridge:** Every PDF in `/evidence/` has a matching `.txt` sidecar.
- **Registry:** `claims.json` maps every `claim_id` to at least one evidence URI.
- **Integrity Seal:** `site_manifest.json.asc` is PGP-signed and current.
- **Open Access:** CORS headers enabled for `/evidence/`.
- **Audit Portal:** `/query/` is live and serves the Forensic Manifesto prompt.

v1.6 Compliant (superset of v1.5)

- All v1.5 requirements above.
- **Vault Resume Format:** Narrative Layer uses dense-stub presentation with `REF: [claim_id]` anchors.
- **Universal Schema:** `claims.json` uses the universal actor/claim/evidence structure.
- **Verification Strength:** Every claim declares `verification_strength.self_declared` with an honest rationale.
- **External Verification:** High-stakes claims include `external_verification` links to public records where available.

13. Future Direction

The W3C Verifiable Credentials standard and digital wallet infrastructure will eventually provide institutional verification for many credential types. The Kerekes Handshake is designed for the gap that institutional infrastructure will not fill: self-sovereign, non-credentialed accomplishments that exist only in primary artifacts.

When VC wallets become broadly adopted, the SHA-256 hashes in the Handshake manifest are already in a compatible format for export. The two approaches are complementary, not competing.

Specialized AI agents — audio analysis, vision models capable of inspecting Photoshop layer stacks, video analysis of project files — will extend the verification reach of the protocol into domains where text-based AI is currently limited. The schema is designed to accommodate these evidence types without modification.

> This protocol is one person's attempt to solve a specific problem. It is deliberately simple, deliberately open, and deliberately honest about what it cannot do. If you find it useful, fork it. If you find the limits unacceptable, improve it. Feedback is more valuable than agreement.
>
> *github.com/jeffreykerekes/kerekes-handshake — Licensed CC BY-SA 4.0*

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PGP Fingerprint: `D39E 4ACE A4FE 3E6B 547F 58C4 6174 3446 DFA7 D48F`

Public Key: `https://jeffreykerekes.com/verify/pubkey.txt`

Reference Implementation: `https://jeffreykerekes.com`

Closing Note

The problem this protocol addresses is not new. People have been overstating their credentials since credentials existed. The apostille system, professional licensing boards, and background check services exist because self-assertion is insufficient for high-stakes decisions.

What has changed is that AI has made narrative optimization nearly free, which lowers the cost of plausible misrepresentation and raises the value of anything that makes evidence directly inspectable. This protocol is one approach to that problem for a specific class of claims — those supported by primary artifacts that an actor controls and can serve publicly.

It works better for some use cases than others. It is most useful where public records exist to create independent verification anchors. It is least useful where the only evidence is self-authored. It is honest about that distinction.

Whether it is useful enough to be worth the effort of implementation is a question each actor has to answer for their own situation. The reference implementation suggests it is — but one data point is not a pattern.

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End of Whitepaper — Kerekes Handshake v1.6.1